Article

Medical Insurance Cost Prediction using Machine Learning

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Abstract: Machine learning techniques are being increasingly applied to predict healthcare costs and improve resource allocation in the medical industry. This study compares the performance of various machine learning algorithms, including linear regression, ridge regression, lasso regression, k-nearest neighbors regression, support vector regression, decision trees, random forests, and XGBoost, in predicting medical insurance costs using a dataset of 1,338 instances from the Kaggle. The algorithms are evaluated based on metrics such as R-squared, root mean squared error (RMSE), Mean squared Error (MSE), Mean Absolute Error (MAE). Linear regression and its regularized variants achieve reasonable accuracy, but more advanced algorithms like XGBoost, random forests, and decision trees, when optimized using grid search, outperform them. Random Forest Regressor exhibits the highest accuracy, followed by XGB Regressor and decision trees. The study highlights the value of machine learning for strategic health care management, especially in predicting insurance costs, and highlights the importance of comprehensive performance by Error Rates, Accuracies.

Keywords: Linear Regression ; Lasso Regression; Ridge Regression **;** Decision Tree Regressor ; Random Forest Regressor ; support vector Regressor (SVR);

Introduction

Machine learning projects improve patient experience and care services in the healthcare industry. Health care faces many challenges, and the cost is an issue for insurers. Medical bills are one of the most frequent expenses that arise over a person's lifetime. Based on studies, factors such as body mass index (BMI), age, smoking, and others are associated with higher medical costs for individuals. Estimating the health costs associated with conditions such as obesity is important to help develop cost-effective prevention strategies.

Organizations have achieved better patient experience through machine learning. However, despite the potential benefits of advanced machine learning in the forecasting industry, insurers still rely heavily on traditional linear regression models to manage patient populations may be due to methods a slow uptake is partly due to unknown effects and difficulty in interpreting results.

Recent literature has described methods for estimating physical health expenditure using different methodologies. The study used machine learning to determine the healthcare costs associated with conditions such as congestive heart failure, hospital readmissions, spinal surgery, and predictive analytics of chronic diseases have the potential to cost improve accountability, enable preventive care, and encourage better resource allocation in health care.

Literature Review:

Previously, the regression techniques like Linear Regression, Lasso Regression, Ridge Regression is used for Predicting the medical insurance cost. However, this models rely on mapping the features and target variable i.e., they try to find the linear relationship between features and target variable. However, linearity does not hold true in all cases. There are more flexible models such as Decision Trees, Random Forests, XG Boost. Additionally SVR and KNN shown good results in capturing non-relationship in data. In recent years, conversion of regression into classification become popular allowing classification algorithms like SVM and Ensemble Methods.

The literature emphasizes the continued study of machine learning techniques and information exploration, especially in the area of claim prediction. In 2019, Jessica Pesantez-Narvaez conducted a study in which she compared the predictive power of XGBoost and logistic regression techniques for auto insurance claims. Remarkably, because logistic regression is more interpretable and predictable than XGBoost, the results preferred it. Even though logistic regression proved to be more effective in some situations, more research is required to fully comprehend the subtleties of claim prediction in various datasets and contexts.[1]

There is a lot of composition on the chance of medical care and its advantages. A pre-portion what's more, risk-pooling framework, medical care deals with clinical costs related with disorder. Different supporting plans give moving degrees of client choice and government control, for instance, single and various medical care pools. Plus, a variety of clinical benefits needs and tendencies are obliged by various medical care plans, including PPOs, HDHPs with HSAs, and HMOs (prosperity upkeep affiliations). The meaning of medical care in progressing financial security and thriving is highlighted by understanding its benefits, which consolidate consideration against fundamental sickness, assurance from rising clinical expenses and crisis admittance to medical services [2]

The survey of the writing plunges into various investigations on the utilization of AI procedures and guarantee forecast. There are numerous methods for anticipating insurance claims, as exemplified by studies such as Jessica's that used telematics data to predict insurance claims and Ranjodh Singh's method for figuring out how much it will cost to fix cars based on pictures. Moreover, Oskar Sucki's exploration on client stir forecast and Muhammad rFauzan's examination of AI techniques feature the worth of cutting edge investigation in handling issues in the insurance sector. Yet, these examinations often run into issues like missing information and one-sided philosophy, which recommends that more solid and ethically sound techniques for guarantee expectation are required [3]

The use of AI calculations to figure medical charges and clinical costs is a subject shrouded exhaustively in the writing. For more exact expectations, studies recommend that utilizing this calculations like Arbitrary Woodland Relapse (RFR) and Outrageous Slope Helping (XGBoost). The goal is to make predictions more accurate and efficient. principal objective of investigating ensemble methods and multiple regression models. In an work to create more exact assessments for medical coverage claims, specialists are still exploring novel strategies for assessing protection costs, even despite predisposition and missing information[4]

The review of the literature concentrates on the research on machine learning and knowledge discovery in relation to the prediction of insurance costs. Research demonstrating variety of methods in claim prediction include Jessica's comparison of XGBoost and regression techniques and Ranjodh Singh's system for estimating repair costs. But problems occurs like incomplete data and other methodologies like skewed methodology still exist, this is why researchers are looking into more sophisticated approaches like XGBoost to deal with missing values. Further, research on the identification of fraud claims tells the significance of sophisticated statistical techniques and machine learning algorithms in getting the prediction of insurance claims.[5]

The literature review examines the use of information and machine learning techniques to predict insurance claims. Research by Jessica Pesantez-Narvaez and partners exhibited how well XGBoost and calculated relapse work to assess fix costs and anticipate mishap claims. Notwithstanding, this issues like inadequate information and slanted philosophy make it important to research for more strategies for guarantee expectation. Researchers are focusing on deep neural networks, machine learning algorithms, and advanced statistical methods in order to better predict health insurance claims.[6]

The study emphasizes the significance of accurately predicting future healthcare costs. It specifies various techniques like managed learning and profound realizing, which can help in making these forecasts. Utilizing itemized clinical information from things like electronic wellbeing records and health care coverage claims is viewed as vital for controlling medical services costs actually. When estimating healthcare costs, the research also demonstrates that it is preferable to employ objective techniques like machine learning rather than making educated guesses. Generally speaking, the review features that it is so fundamental to anticipate costs precisely for overseeing medical services well, and how AI procedures can truly assist with that.[7]

The literature offers a variety of strategies to deal with problems in a number of fields, such as market forecasting and insurance. Relational Random Forests (RRFs) provide a new approach to relational data handling that incorporates dynamic propositionalization and overcomes the drawbacks of FORF and other previous methods. While managing information that is imbalanced, outfit strategies, similar to a Gathering Arbitrary Woods Calculation for protection investigation, perform better compared to additional regular methodologies like SVM. Insurance contract assessing includes helping capital apportioning while at the same time checking the risks of obligation and projected benefits. The KGB method for electricity price forecasting and other models that incorporate intraday data and the effects of renewable energy enhance the risk premium calculation. Relapse and order calculations in land utilize different elements to figure home costs with a serious level of precision.[8]

Using AI and machine learning models to predict medical insurance installments is one of ongoing exploration. For insurance cost determining, Kaushik K. and, Bhardwaj proposed a mental registering methodology that utilizes counterfeit brain organizations, while Jessica Pesantez-Narvaez contrasts XGBoost and strategic relapse for mishap guarantee expectation. Utilizing photos of damaged cars, Ranjodh Singh's system calculates repair costs, improving the processing of claims. M. Nandhini and G. Kowshalya demonstrate the superiority of Random Forest by using classifiers to forecast fraud claims. These studies highlight the potential of machine learning to enhance insurance processes, including fraud detection and premium estimation.[9]

The literature on healthcare cost prediction uses a variety of approaches, ranging from sophisticated machine learning techniques to conventional statistical models. When predicting future costs, researchers emphasize using historical cost features—a technique known as "cost on cost prediction." Systematic reviews demonstrate the superiority of this method over models based on clinical data. Current research explores various input features and learning strategies to improve prediction precision. In one study, cost on cost prediction is directly compared with alternative methods using real-world health insurance data, which includes demographics, clinical information, and claims from pharmacies and doctors. Researchers try to find the best methods for guiding healthcare policy and financing decisions by analyzing different models.[10]

4. Proposed Model

Objective:

The primary Objective of this Medical Insurance Cost Prediction is to help Organizations predict Accurate Predictions to set appropriate premiums for policy holders.

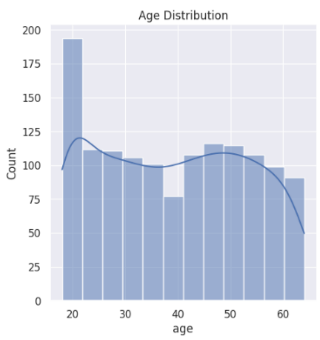
Research Design

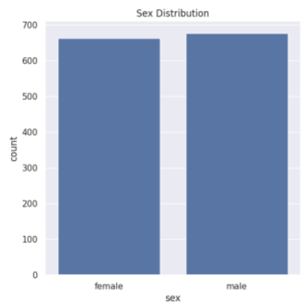
Designing a research study for Medical Insurance Cost Price Prediction using machine learning involves several key components and considerations. Below, I outline a research design that encompasses these essential elements:

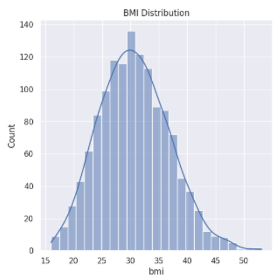
1. Data Collection
2. Data Preprocessing
3. Data Splitting
4. Model Selection
5. Model Training

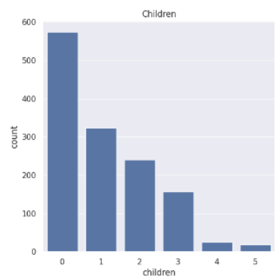
Data Preparation:

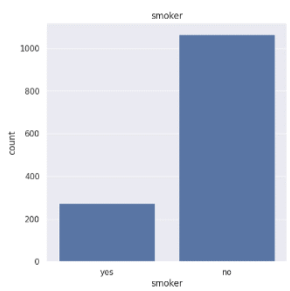
Load the dataset containing medical insurance cost data. Explore the data to understand its structure, missing values, and distribution. Preprocess the data then handling missing values i.e., null values, encoding categorical features, and scaling numerical features.

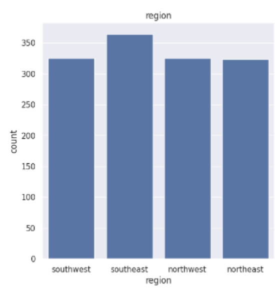


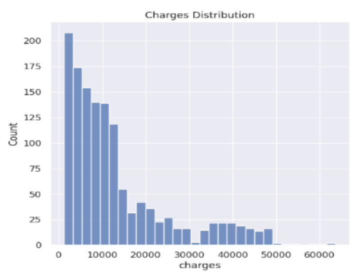












Feature Selection:

Identify relevant features that may impact medical insurance costs. Use domain knowledge or feature importance techniques to select important features.

The features in the dataset are age, sex, Body mass Index(bmi), children, smoker, region

Data Splitting:

Data splitting is a common practice in data analysis and machine learning. It involves the process of dividing a dataset into multiple subsets and this is used to train, test and validate the machine learning models. These are essential for assessing the model's performance, preventing overfitting, and making decisions.

Training Set: To train the machine learning model, this section of the dataset is used. The model uses this portion of the data to identify trends, connections, and correlations between the input features and the target variable. Typically, the training set comprises 70–80% of the total data, making it larger than the test set.

Testing Set: This subset is kept separate from the training data and is used to evaluate the model's performance and generalization. The model has never seen the data in the test set during training. It helps you estimate how well your model will perform on new, unseen data. Typically, the test set contains around 20-30% of the data, but, again, this can vary.

Model Selection:

Model selection is the process of choosing the most appropriate machine learning or statistical model to solve a specific problem. Model selection involves considering various factors, including the dataset we have and the type of problem to solve. Generally there are two types (i.e., classification or regression) . so, in order the model that best fits our dataset is Regression.

Regression has many models

Linear Regression:

A simple linear model that assumes a linear relationship between features and target.

The linear regression model performs good in predicting medical costs with an accuracy of 84.70%. The linear relationship between independent variables and insurance costs is well captured by it. Its precision and recall are 89.42% show that it can accurate to identify cases with high and low insurance costs. The F1 score 84.01% indicates that the model is dependable in identifying both price and cheap insurance plans because it creates a fair balance among recall and precision. The model is a useful for insurance agents and policymakers due to its simplicity and interpretability.

```
LinearRegression Classification Metrics:
Accuracy: 0.8470149253731343
Precision: 0.8942387421865033
Recall: 0.8470149253731343
F1 Score: 0.8400662183654422
```

Lasso Regression: Adds L1 regularization to linear regression, encouraging sparsity in feature coefficients.

The Lasso regression model performs similarly to linear regression. It produces a sparse model with less complexity by placing a penalty on its size of the regression coefficients. The model's high precision and recall values are maintained in spite of this regularization, taking its efficiency in precision categorizing insurance costs. The model's capacity to strike a balance between recall and precision is demonstrated by its F1 score of 84.01%, which makes it a best choice for predicting cost

```
Lasso Classification Metrics:
Accuracy: 0.8470149253731343
Precision: 0.8942387421865033
Recall: 0.8470149253731343
F1 Score: 0.8400662183654422
```

Ridge Regression:

Adds L2 regularization to linear regression, preventing overfitting.

The ridge regression performs similarly to linear and Lasso regression. The model improves its capacity for generalization by mitigating multicollinearity and overfitting by attaching a penalty term to the regression coefficients. precision and recall values are 89.42% demonstrate how well the model can differentiate between high and low insurance costs. Furthermore, the F1 score of 84.01% indicates a well-balanced combination of recall and precision, emphasizing the model's dependability in cost estimation assignments.

```
Ridge Classification Metrics:
Accuracy: 0.8470149253731343
Precision: 0.8942387421865033
Recall: 0.8470149253731343
F1 Score: 0.8400662183654422
```

Decision Tree Regressor:

A non-linear model that splits data based on feature thresholds.

The decision tree regressor performs better than models based on linear regression with an accuracy of 87.69%. By using hierarchical tree structures to divide the features, it is able to efficiently capture interactions and nonlinear relationships between variables.precision and recall values are 87.78% show that how well the model can categorize insurance costs. Moreover, the F1 score of 87.51% highlights the model's resilience in cost prediction tasks by showing a balanced trade-off between precision and recall.

```
DecisionTreeRegressor Classification Metrics:
Accuracy: 0.8768656716417911
Precision: 0.8777910922335856
Recall: 0.8768656716417911
F1 Score: 0.8750538479081672
```

Random Forest Regressor:

An ensemble of decision trees that reduces overfitting.

The random forest regressor outperforms all the models with an accuracy of 92.54%. The ensemble model increases prediction accuracy and decreases overfitting by combining several decision trees.precision and recall values are 92.35% tells how well the model can classify insurance costs. In addition, the F1 score is 92.03% highlights the model's robustness and dependability in cost estimation tasks by indicating a pleasing balance between precision and recall.

```
RandomForestRegressor Classification Metrics:
Accuracy: 0.9253731343283582
Precision: 0.9234528371655237
Recall: 0.9253731343283582
F1 Score: 0.9202896566231469
```

Support Vector Regressor (SVR):

Uses to find support vectors to find a hyperplane that best fits.

The support vector regression does moderately well in forecasting insurance prices with an accuracy of 79.85%. The seems to reduce cost estimating mistakes by building a hyperplane that maximizes the margin between data points. The model can correctly classify cases of high and low insurance expenses.The moderate degree of precision across all metrics shown by the F1 score of 70.90%, however, data points to possible areas for improvement.

```
SVR Classification Metrics:
Accuracy: 0.7985074626865671
Precision: 0.8391067052795723
Recall: 0.7985074626865671
F1 Score: 0.7090481203938812
```

K-Nearest Neighbors Regressor:

Predicts based on the average of k nearest neighbors.

At 76.49% accuracy, the k-nearest neighbors regressor performs moderately. Based on similar instances in the feature space, the model estimates insurance costs by averaging the values of the k nearest data points. The 72.05% precision and recall figures show how well the model can identify cases with high and low insurance costs. Though there may be room for improvement, the F1 score of 70.89% indicates a moderate degree of precision across both measurements.

```
KNeighborsRegressor Classification Metrics:
Accuracy: 0.7649253731343284
Precision: 0.7205273631840796
Recall: 0.7649253731343284
F1 Score: 0.7089061272584446
```

XG Boost Regressor:
A gradient boosting algorithm that combines weak learners.

XG Boost regressor performs competitively, achieving 88.43% accuracy. XG Boost is an ensemble learning technique that produces reliable and accurate estimates of insurance costs by aggregating the predictions of several weak learners (decision trees). The 91.35% precision and recall figures demonstrate how well the model can classify insurance costs. Furthermore, the F1 score of 89.58% indicates a good trade-off between recall and precision, highlighting the model's dependability and resilience in cost estimating tasks.

```
XGBRegressor Classification Metrics:
Accuracy: 0.8843283582089553
Precision: 0.9134789249840843
Recall: 0.8843283582089553
F1 Score: 0.8958080352174503
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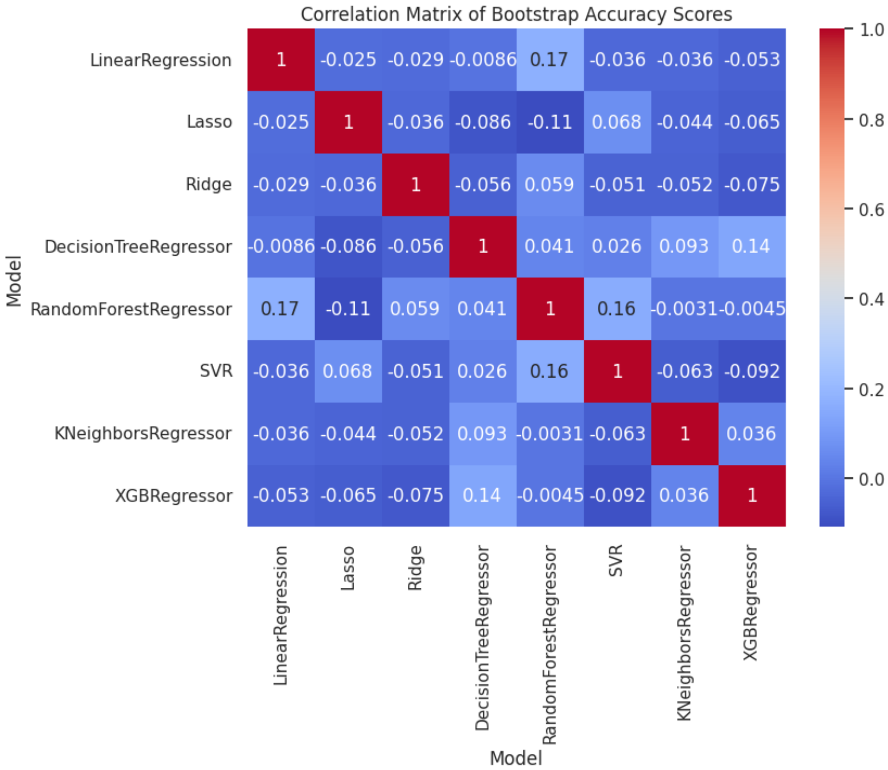
Correlation Matrix for All Models:



Results:

MSE of All Models:

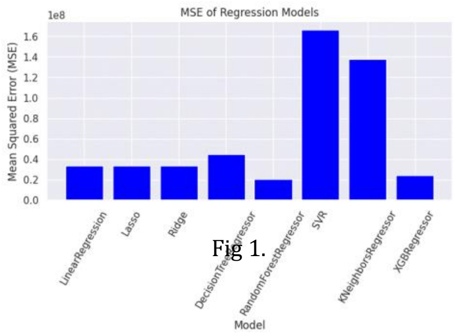


Fig 1.

The above fig 1.1 showcases the MSE error rates for all the regression modelswhere random Forest Regressor has the low error rate and Support vector Regressor has the highest Error Rate. It is concluded that the model with lowest Error Rate is the best

MAE of All Models:

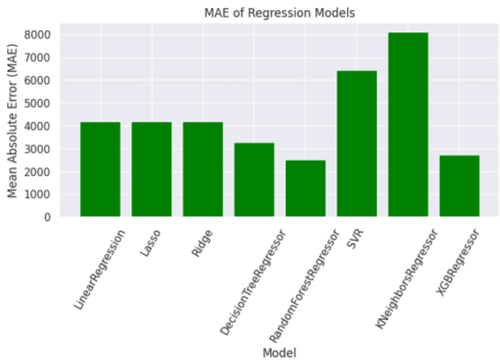


Fig 1.2

The above fig 1.2 showcases the MAE error rates for all the regression models from 1000 to 8000 where random Forest Regressor has the low error rate and K Neighbors Regressor has the highest Error Rate.

RMSE of All Models:

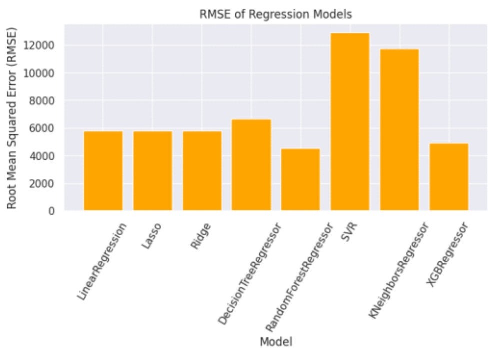


Fig 1.3

The above fig 1.3 showcases the MAE error rates for all the regression models from 1000 to 8000 where random Forest Regressor has the low error rate and K Neighbors Regressor has the highest Error Rate.

R2 Score:

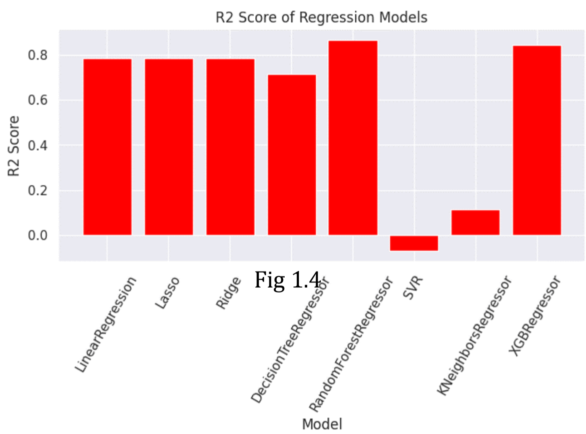
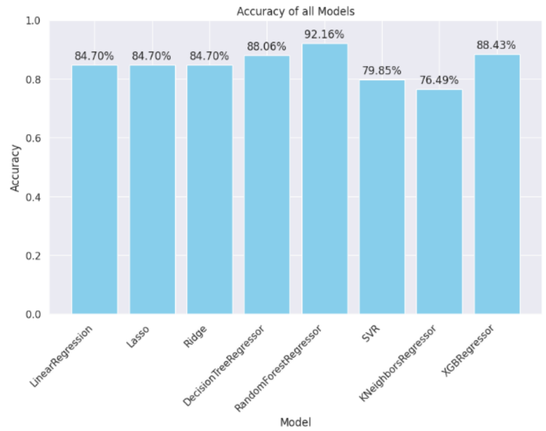


Fig 1.4

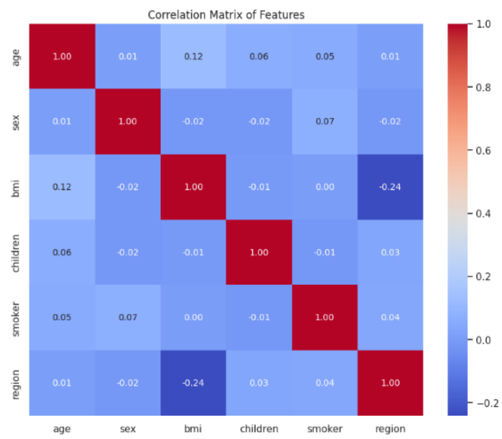
The above fig 1.4 showcases the R2 Score for all the regression models from 1000 to 8000 where random Forest Regressor has the highest R2 Score and whereas SVR performs low R2 Score.

Final Accuracy:

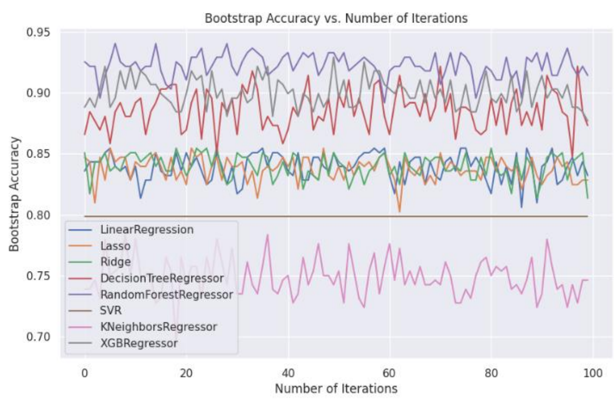


Discussion:

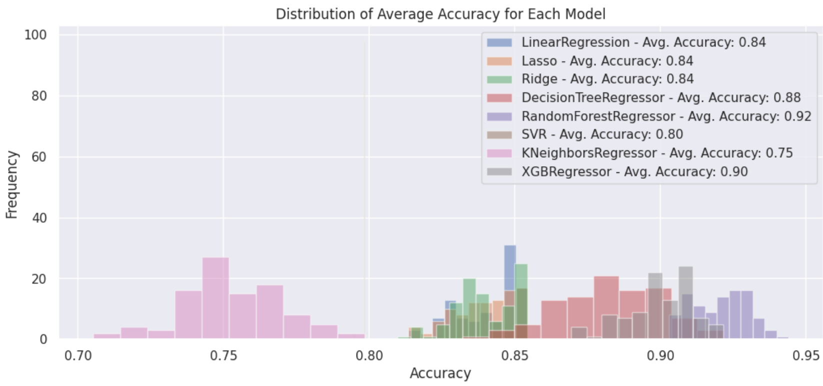
A correlation matrix was constructed to examine the relationship between the attributes and the objective variables. This analysis helps to identify potential multicollinearity problems and informs selection methods, ultimately improving the overall performance of the models



Random forest regressor consistently outperforms other models when regression models are evaluated using performance metrics like mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), and R2 score. Furthermore, the regression was modelled as a classification function to revalidate the model's classification accuracy. Then the random forest regressor was performing well, obtaining an accuracy of 92.16%. This suggests that the random forest algorithm not only works well at predicting ongoing changes, but it also demonstrates its ability to solve classification problems.



An approach known as bootstrapping remodelling was employed with the model's correctness. By minimizing its impact of sample variability, this technique tells that it estimates of sample performance that are more trustworthy. A rigorous evaluation of the models was accomplished by bootstrapping the accuracy ratings, guaranteeing a more representative evaluation of the models' predictive potential.



By incorporating bootstrapping, average accuracy calculations, and insights from correlation matrix analysis into performance measures, the superiority of the random forest regressor becomes more apparent, strengthening its impact.

Conclusion:

The final conclusion is it draws from a variety of techniques including bootstrapping, classification accuracy and it's performance simulation, and correlation matrix testing confirm that the random forest regressor is a highly effective regression model component as it outperforms well. Its outstanding flexibility in classification tasks and its consistent supremacy in forecasting continuous change, as evidenced by its excellent accuracy of 92.16%, solidify its place as a strong and adaptable approach at emphasize machine learning.

Furthermore, we were able to determine the predictive capability of each model in greater detail and with greater assurance thanks to the use of bootstrapping techniques in the calculation of these scores to evaluate accuracy.
In particular, our results demonstrate not only the effectiveness of the random forest regressor but also the value of a multi approach to model analysis, with various methods emphasizing a full ability the model's operational expenses. Using this method not only makes our findings more reliable, but it also offers insightful information about predictive models and how to employ them in future studies.